

# Engineering Physics (2025)

Course code 25PY101

## Module 2 Unit 3: Lasers

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Laser

Optical fibre

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# Laser

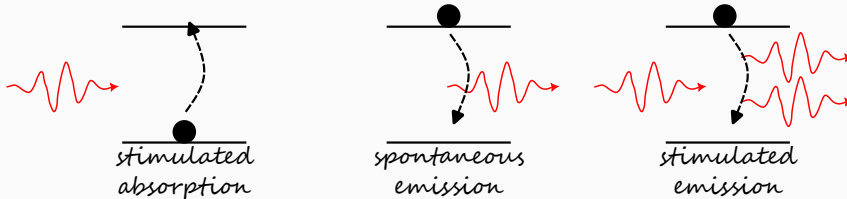
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Laser

Optical fibre

# Laser – Three Quantum Processes

- L.A.S.E.R is acronym for Light Amplication by Stimulated Emission of Radiation.
- Einstein predicted the possibility of lasing in 1916 by introducing concept of stimulated emission.



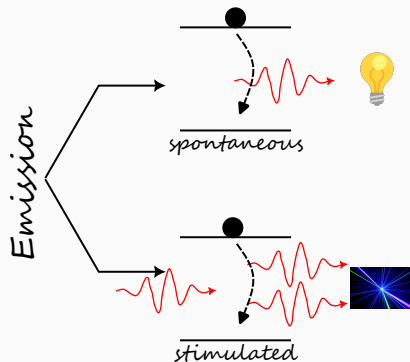
- Light interacts with matter by three quantum processes
  - Stimulated absorption
  - Spontaneous emission
  - Stimulated emission

# Lasing action

## Definition

Lasing action is defined as the condition when the stimulated emission dominates the spontaneous emission process.

- Emission of radiation when the electron makes a transition from higher energy level to lower energy level can be either
  - spontaneous
  - stimulated
- In normal light 💡, the spontaneous emission dominates stimulated emission.
- In laser light 🌈, the stimulated emission dominates spontaneous emission.
- Einstein's theory of light matter interaction requires **stimulated emission** and **high photon density** to achieve lasing action.



# Laser – Conditions for lasing action

There are three conditions to be satisfied for lasing action –

1. Population inversion
2. Metastable state
3. Radiation confinement



# Thermal equilibrium and normal state

## Definition

Maxwell-Boltzmann statistics says the probability of occupying higher energy level is lower than the probability of occupying lower energy level. The probability of occupying energy level  $E$  is proportional to

$$P_{MB} = \exp\left(-\frac{E}{k_B T}\right)$$



- Under the conditions of thermal equilibrium, the occupancy of energy levels is governed by Maxwell Boltzmann statistics.

## Problem

*Find the ratio of populations of two states in a He–Ne laser that produces light of wavelength 6328 Å at 27 °C.*



*normal state*

# Population inversion and inverted state

## Definition

Population inversion is defined as the condition when the number of electrons in the higher energy level is more than the number of electrons in the lower energy level.

- Under the conditions of thermal equilibrium, the occupancy of energy levels is governed by Maxwell Boltzmann statistics.



## Problem

A 10 mW *He-Ne laser* has efficiency of 1 %.

*Assume that all input energy is utilized in pumping the atoms from ground state to the excited state, which is 20 eV above the ground state. Find how many atoms are promoted to the excited state per second.*



*inverted state*

## Definition

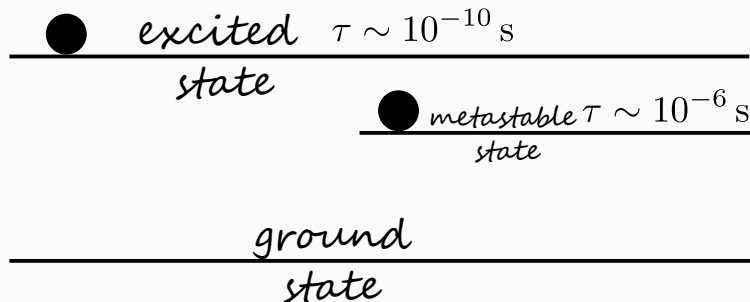
Inverted state is defined as the state attained after population inversion.

# Metastable state

## Definition

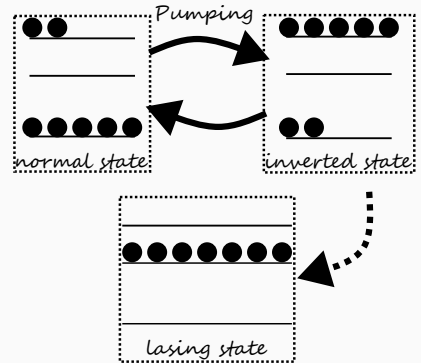
Metastable state is an excited state with an unusually long life time.

- Excited states have a very low lifetime of the order of 10 ns.
- However, some excited states can have larger lifetimes due to various reasons. These are called metastable states.
- These states lie in between excited states and ground states both in terms of lifetime and energy.
- The metastable state lifetime is of the order of  $10^{-3}$  s to  $10^{-6}$  s.



## Metastable state $\Rightarrow$ lasing state

- If the lifetime of electron in metastable state is  $10^{-3}$  s to  $10^{-6}$  s, then the electron stays nearly  $10^2$  to  $10^5$  times longer in the metastable state than in the excited state.
- **Analogy:** To understand the scale difference of lifetimes, if we consider lifetime of excited state as 1 s, then lifetime of metastable state will be as 100 s to 100 000 s i.e. 2 minute to 1 day.
- If the pumping is continued, due to difference in time scale, spontaneous emission from the excited state significantly populates the metastable state. Such a condition for is called the **lasing state**.



**Definition**

Radiation confinement is the one of the conditions to achieve lasing action, where the photons are confined to a region in space and time.

# Laser – components

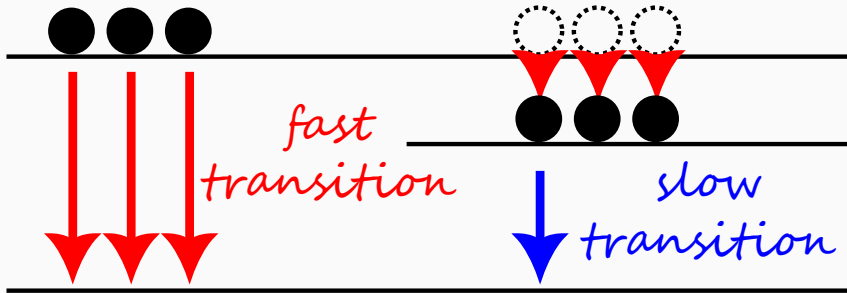
There are three components to any laser –

1. Active medium
2. Optical pump
3. Optical resonant cavity

# Active medium

## Definition

Active medium is the physical medium where lasing action takes place. The medium hosts the metastable state that has larger lifetime so that population inversion can be sustained.

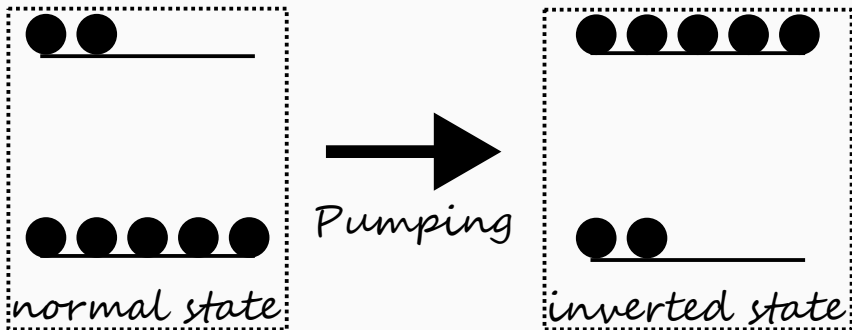


# Optical pump

## Definition

Optical pump is the external agent that transforms the active medium from normal state to inverted state.

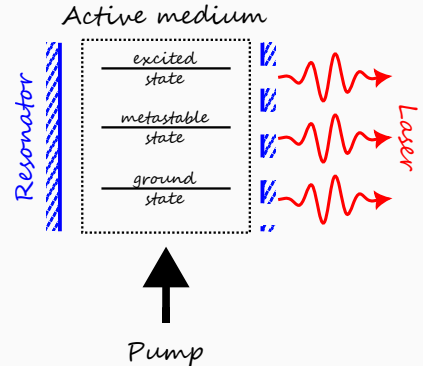
- Optical pump provides the energy to raise the electrons from ground state to the excited so that population inversion takes place.





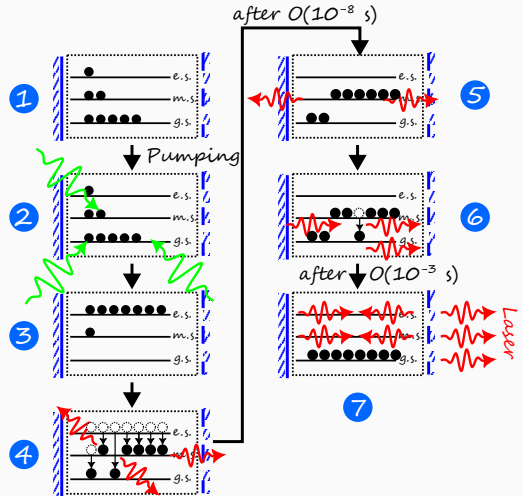
# Optical resonator

- In order to confine the radiation within the active medium, the reflection property of light is used.
- A pair of optically flat/plane mirrors enclose the active medium so that the photons get confined.
- In order to extract the photons from a single direction, one mirror is made fully reflecting whereas second mirror is partially reflecting. This arrangement is called optical resonant cavity or simply optical resonator.
- The commonly used optical resonator is called Fabry-Perot interferometer.



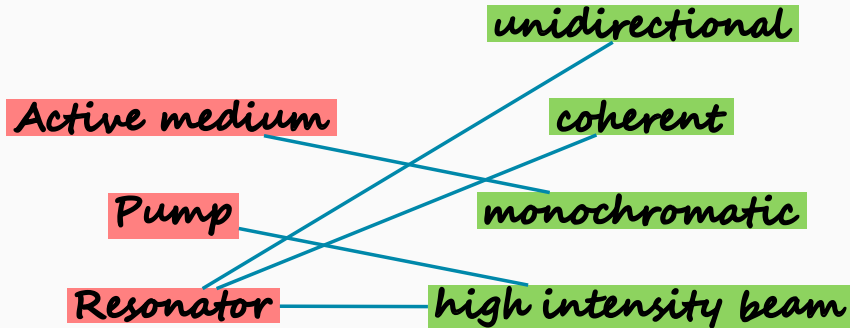
# Lasing action

1. Normal state
2. Stimulated absorption
3. Inverted state
4. Spontaneous emission
  - most photons along the resonator axis are lost.
  - some photons along the resonator axis are trapped.
  - photons of other color are lost or reabsorbed.
5. Lasing state
6. Stimulated emission
  - Emitted photons further stimulate other transitions leading to **gain/amplification**.



## Lasing action (contd...)

- Lasing action occurs because
  - multiplication of photons leads to high intense beam.
  - a fraction of photons escape the resonator due to partially reflected mirror.
- Lasing action leads to

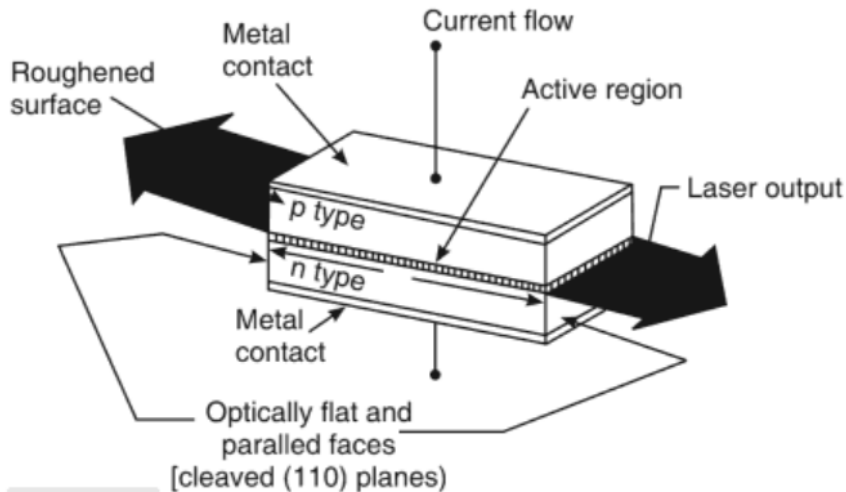


## Laser – characteristics

| Laser light  | Ordinary light  |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Unidirectional                                                                                | Non-directional                                                                                    |
| Coherent                                                                                      | Incoherent                                                                                         |
| Monochromatic                                                                                 | Non-monochromatic                                                                                  |
| High intensity                                                                                | Low intensity                                                                                      |
| Low pulse width                                                                               | High pulse width                                                                                   |

- Coherent means photons in the beam have same phase
- Monochromatic beam consists of photons of single color/wavelength.

## Diode laser – Construction

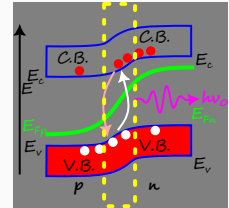
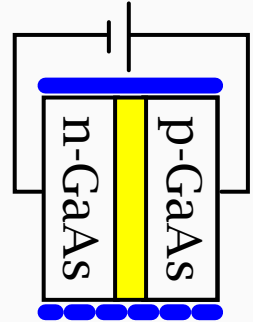


**Fig. 44.24:**

Schematic of homojunction diode laser

# Diode laser – working principle

- In a diode laser
  - Depletion region is the active medium,
  - Forward biasing is the pump, and
  - Polished faces of the GaAs crystal are the optical resonator.
- Since the forward biasing populates the electrons in the n region and holes in the p region, when they meet at the depletion region, the active medium is already in the lasing state.
- Therefore, there is no need for a meta-stable state.



## Diode laser – applications

- Industrial precision cutting
- Lasik surgery
- LIDAR (Light Detection and Ranging) technology for distance measurement
- Optical fibre communication technology

# Optical fibre

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